

Safety data sheet

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BASF Safety data sheet
Date / Revised: 07.06.2019
Product: **Trilon® B Powder**

Version: 5.0

(30043442/SDS_GEN_TH/EN)

Date of print 16.01.2023

1. Substance/preparation and manufacturer/supplier identification

Trilon® B Powder

Use: complexing agents for the chemical industry

Manufacturer/supplier:

BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-57579
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:

International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

Acute toxicity: Cat. 4 (Inhalation - dust)

Acute toxicity: Cat. 4 (oral)

Serious eye damage/eye irritation: Cat. 1

Specific target organ toxicity — repeated exposure (Respiratory system): Cat. 2 (by inhalation)

Label elements and precautionary statement:

Pictogram:



Signal Word:
 Danger

Hazard Statement:

H318 Causes serious eye damage.
 H332 Harmful if inhaled.
 H302 Harmful if swallowed.
 H373 May cause damage to organs (Respiratory system) through prolonged or repeated exposure (inhalation).

Precautionary Statements (Prevention):

P271 Use only outdoors or in a well-ventilated area.
 P280 Wear eye/face protection.
 P261 Avoid breathing dust.
 P260 Do not breathe dust/gas/mist/vapours.
 P270 Do not eat, drink or smoke when using this product.
 P264 Wash with plenty of water and soap thoroughly after handling.

Precautionary Statements (Response):

P310 Immediately call a POISON CENTER or doctor/physician.
 P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
 P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
 P330 Rinse mouth.

Precautionary Statements (Disposal):

P501 Dispose of contents/container to hazardous or special waste collection point.

Other hazards which do not result in classification:

The product is under certain conditions capable of dust explosion.

3. Composition/information on ingredients

Chemical nature

tetrasodium ethylenediaminetetraacetate
 CAS Number: 64-02-8

Hazardous ingredients

| tetrasodium ethylenediaminetetraacetate

Content (W/W): >= 50 % - <= 100 %
 CAS Number: 64-02-8

Acute Tox.: Cat. 4 (Inhalation - dust)
 Acute Tox.: Cat. 4 (oral)
 Eye Dam./Irrit.: Cat. 1
 STOT RE (Respiratory system): Cat. 2 (by inhalation)

| sodium hydroxide

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Content (W/W): $\geq 0\%$ - $< 3\%$ CAS Number: 1310-73-2	Met. Corr.: Cat. 1 Skin Corr./Irrit.: Cat. 1A Eye Dam./Irrit.: Cat. 1
trisodium nitrilotriacetate Content (W/W): $\geq 0\%$ - $< 5\%$ CAS Number: 5064-31-3	Acute Tox.: Cat. 4 (oral) Eye Dam./Irrit.: Cat. 2A Carc.: Cat. 2 Aquatic Acute: Cat. 3
Acetic acid, hydroxy-, monosodium salt Content (W/W): $\geq 0\%$ - $< 5\%$ CAS Number: 2836-32-0	Skin Corr./Irrit.: Cat. 2 Eye Dam./Irrit.: Cat. 1

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:

Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: respiratory disorders, corneal injury, gastrointestinal complaints, irritation of the mucous membranes

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

dry powder, foam

Unsuitable extinguishing media for safety reasons:

carbon dioxide

Additional information:

Avoid whirling up the material/product because of the danger of dust explosion.

Specific hazards:

harmful vapours, carbon oxides, nitrogen oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations. Dusty conditions may ignite explosively in the presence of an ignition source causing flash fire.

6. Accidental Release Measures

Personal precautions:

Avoid dust formation. Use personal protective clothing. Information regarding personal protective measures, see section 8.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For small amounts: Pick up with suitable appliance and dispose of.

For large amounts: Contain with dust binding material and dispose of.

Avoid raising dust. Dispose of absorbed material in accordance with regulations.

Additional information: Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

7. Handling and Storage

Handling

Provide exhaust ventilation. Avoid inhalation of dusts.

Protection against fire and explosion:

Avoid dust formation. The product is capable of dust explosion. Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame.

Storage

Suitable materials for containers: Low density polyethylene (LDPE), glass, Paper/Fibreboard, High density polyethylene (HDPE)

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Protect from temperatures below: -20 °C

Protect from temperatures above: 70 °C

8. Exposure controls and personal protection

Components with occupational exposure limits

sodium hydroxide, 1310-73-2;
CLV 2 mg/m³ (ACGIHTLV)
TWA value 2 mg/m³ (OEL (TH))

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):

e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is recommended. Keep away from food, drink and animal feeding stuffs. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form: powder
Colour: white
Odour: product specific

pH value: approx. 10.5 - 12.5 (DIN 19268)
(10 g/l, 23 °C)

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Melting point:	The substance / product decomposes therefore not determined.	
decomposition point:	> 150 °C Literature data. The substance / product decomposes.	
Flash point:	Study scientifically not justified.	
Evaporation rate:	The product is a non-volatile solid.	
Flammability (solid/gas):	not highly flammable	(VDI 2263, sheet 1, 1.1)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Ignition temperature:	> 200 °C	(DIN 51794)
Thermal decomposition:	> 300 °C	
Self heating ability:	It is not a substance capable of spontaneous heating.	
Minimum ignition energy:	> 4 J The product is capable of dust explosion.	(VDI 2263, sheet 1, 2.1.1)
Explosion hazard:	Product is not explosive, however a dust explosion could result from an air / dust mixture.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	6 hPa (approx. 25 °C) contains water, Literature data.	(measured)
Relative density:	1.67 (20 °C) Literature data.	
Bulk density:	620 - 760 kg/m ³	
Solubility in water:	approx. 750 g/l	
Solubility (qualitative) solvent(s):	polar solvents soluble	
Partitioning coefficient n-octanol/water (log Pow):	-13 (20 °C)	

:

Adsorption to solid soil phase is not expected. Under environmental conditions, the substance will almost completely be in its charged form.

Viscosity, dynamic:

Study scientifically not justified.

10. Stability and Reactivity

Conditions to avoid:

Avoid humidity. Avoid dust formation.

Thermal decomposition: > 300 °C

Substances to avoid:

amphoteric metals, light metals

Corrosion to metals: Corrodes metals in the presence of water or moisture.

Hazardous reactions:

Dust explosion hazard.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion. Of moderate toxicity after short-term inhalation.

Experimental/calculated data:

LD50 rat (oral): 1,000 - 2,000 mg/kg (BASF-Test)

LC50 rat (by inhalation): > 1 mg/l (other)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. An aerosol was tested.

LD50 (dermal):

Study scientifically not justified.

Irritation

Assessment of irritating effects:

Not irritating to the skin. May cause severe damage to the eyes.

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (BASF-Test)

Serious eye damage/irritation rabbit: irreversible damage (BASF-Test)

Respiratory/Skin sensitization

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:

In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays.

Carcinogenicity

Assessment of carcinogenicity:

In long-term studies in rats and mice in which the substance was given by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:

Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure):

Assessment of STOT single:

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated inhalation exposure may affect certain organs. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

Not relevant.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. Based on long-term (chronic) toxicity study data, the product is very likely not harmful to aquatic organisms.

Toxicity to fish:

LC50 (96 h) > 100 mg/l, *Lepomis macrochirus* (OPP 72-1 (EPA-Guideline), static)

Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic invertebrates:

EC50 (48 h) > 100 mg/l, *Daphnia magna* (DIN 38412 Part 11, static)

Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants:

EC50 (72 h) > 100 mg/l (growth rate), *Scenedesmus obliquus* (Directive 88/302/EEC, part C, p. 89, static)

Nominal concentration.

Microorganisms/Effect on activated sludge:

EC20 (30 min) > 500 mg/l, activated sludge, domestic (OECD Guideline 209, aquatic)

Nominal concentration. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to fish:

No observed effect concentration (35 d) $\geq$ 36.9 mg/l, *Brachydanio rerio* (OECD Guideline 210, Flow through.)

The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d), 25 mg/l, *Daphnia magna* (OECD Guideline 211, semistatic)

Nominal concentration.

Analogous: Assessment derived from products with similar chemical character.

Soil living organisms:

LC50 (14 d) 156 mg/kg, *Eisenia foetida* (OECD Guideline 207, artificial soil)

Analogous: Assessment derived from products with similar chemical character.

Terrestrial plants:

No observed effect concentration (21 d), terrestrial plants (other)

Analogous: Assessment derived from products with similar chemical character.

Other terrestrial non-mammals:
Study scientifically not justified.

Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Persistence and degradability

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis):
According to structural properties, hydrolysis is not expected/probable.

Sum parameter

Theoretical Oxygen Demand (ThOD): 654 mg/g

Bioaccumulation potential

Bioaccumulation potential:
Bioconcentration factor: approx. 1.8 (28 d), *Lepomis macrochirus*
Does not significantly accumulate in organisms.

Additional information

Other ecotoxicological advice:
Do not release untreated into natural waters.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

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Air transport
IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Hazard determining component(s) for labelling: ETHYLENEDIAMINETETRAACETIC ACID NA4-SALT

Other regulations

16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Vertical lines in the left hand margin indicate an amendment from the previous version.

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